

BREAK INTO AEROSPACE

JAXA (Japan Aerospace Exploration Agency)

Engineering & Research Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

I have what I need. JAXA is one of the most structurally complex organisations in this series — simultaneously a national space agency, a national research and development institution, an aeronautics research centre, and a science institute with a distinct academic hiring tradition — whose three merged founding identities still shape how different parts of the organisation operate today.

CRITICAL STRUCTURAL NOTE BEFORE PART I

JAXA was formed in 2003 through the merger of three institutions: the Institute of Space and Astronautical Science (ISAS), the National Aerospace Laboratory of Japan (NAL), and the National Space Development Agency of Japan (NASDA). It became a National Research and Development Agency in April 2015. These three founding organisations retain distinct identities within JAXA and this matters for interview preparation:

- **ISAS (Sagamihara):** JAXA's science and planetary exploration institute — Japan's equivalent of JPL. It has its own academic hiring system for Professor, Associate Professor, and Assistant Professor positions, governed by the ISAS Advisory Council, with a process closer to a university faculty search than a corporate interview. ISAS's missions include Hayabusa2, MMX, DESTINY+, and BepiColombo.
- **TKSC (Tsukuba Space Center):** JAXA's primary engineering site — satellite operations, HTV-X, ISS Kibo, H3 launch support, and human spaceflight activities are based here.
- **Chofu (Aeronautics Research):** Continues NAL's legacy in aeronautics — hypersonic flight, aviation safety, and next-generation aircraft technology.

Identifying which of these three cultures you are entering is more important than knowing JAXA generically.

PART I: WHAT JAXA ACTUALLY LOOKS FOR

By Level

(New Graduate /)

JAXA's new graduate technical selection runs: entry sheet (ES) → document screening → aptitude test (WEB test) → first through third interviews — a standard Japanese hiring flow. The process is not special or unusual — no English interviews, no special examinations — and is similar to other Japanese companies. The selection is broadly open: technical staff come from diverse backgrounds including mechanical, electrical, electronics, information, communications, control, materials, physics, chemistry, and biology — applicants whose major is not directly space or aerospace-related are explicitly welcomed.

What JAXA screens for at graduate level is specific and characteristically Japanese: interviewers pay attention to how hard students worked in their own academic major and research — the question is whether they have their own specialty. Students who balance academic dedication with extracurricular commitment (sports, music, creative pursuits) are noted positively — JAXA staff are described as people who pursue

their hobbies with the same intensity as their work. This is not a STEM-only filter — it is a curiosity and dedication filter applied through academic formation.

Career Hire (/)

JAXA's career hiring process has shifted to field-led selection () — rather than HR managing selection centrally, the relevant technical group conducts the evaluation because wide-ranging specialist expertise requires accurate assessment by practitioners rather than generalist HR interviewers. The process now runs: document screening + video submission (1st selection) → online 1st, 2nd, and final interviews, all conducted digitally.

The career hiring focus is on immediately deployable specialist expertise: candidates from industry who bring skills JAXA's teams need right now — whether in satellite systems, propulsion, mission analysis, robotics, data science, or materials — are the primary target. Unlike new graduate hiring, career hires are not expected to cover the breadth of JAXA's mission — they are hired for a specific technical capability gap.

ISAS Academic Staff (— Professor / Associate Professor / Assistant Professor)

ISAS positions are selected by the ISAS Advisory Council for Research and Management. The process includes document submission, potential additional document requests, and interviews — with all transportation and accommodation costs borne by the applicant. This is a university-style faculty search, not a corporate hiring process. Publication record, research programme coherence, and the fit between the candidate's scientific agenda and ISAS's current mission portfolio are the primary selection criteria — identical to JPL at the senior level.

Hands-On Experience and Domain Knowledge They Treat as Strong Signals

- **Spacecraft systems engineering** — satellite bus design, GNC, thermal, power, structures. JAXA develops satellites, cargo vehicles, and exploration spacecraft from concept through operations. Systems engineers who have worked through the full development lifecycle on a real spacecraft are directly deployable.
- **H3 and launch vehicle engineering** — LE-9 engine, SRB-3, propulsion systems. The H3 rocket — developed jointly by JAXA and MHI, featuring the LE-9 engine on the first stage and LE-5B-3 on the second — achieved its first successful operational launch in February 2024 after a first-flight failure in March 2023. Propulsion engineers, launch operations engineers, and systems engineers with launch vehicle experience are in demand as the H3 fleet ramps.
- **Planetary science and deep space mission experience.** JAXA's active deep space missions include Hayabusa2 (in transit to asteroid 1998 KY26 for a second target flyby), BepiColombo (Mercury orbit insertion planned November 2026), MMX (Martian Moons eXploration, H3 launch 2026), and DESTINY+ (Phaethon asteroid flyby). Scientists and engineers with asteroid, planetary, or small body experience are specifically relevant to ISAS.
- **Asteroid sample return technology** — curation, sample analysis, re-entry systems. Hayabusa2's sample return from asteroid Ryugu in 2020 is Japan's most celebrated recent scientific achievement, and the technology developed — ion engines, re-entry capsules, sample curation — informs MMX's Phobos sample return design.
- **Lunar exploration engineering** — landers, rovers, in-situ resource utilisation. SLIM (Smart Lander for Investigating Moon) achieved Japan's first successful lunar landing in January 2024, surviving three lunar nights — and JAXA is developing the Lunar Cruiser pressurised rover with Toyota for Artemis, and the LUPEX lunar polar exploration rover with ISRO.
- **ISS operations and human spaceflight systems** — Kibo module, HTV-X. HTV-X1 successfully delivered cargo to the ISS aboard H3-24L in October 2025, and HTV-X is designed to eventually deliver cargo to Lunar Gateway. Human spaceflight engineers with ISS operations or cargo vehicle experience are relevant.
- **Japanese language at professional working level.** JAXA's daily engineering environment — design reviews, internal programme reviews, supplier meetings with MHI, IHI, and NEC — operates in Japanese. English is actively used in international programmes (ESA, NASA, CNES, DLR) but is

not the internal working language.

- **Publication record for ISAS and research-track positions.** For ISAS academic staff and research-track positions, a peer-reviewed publication record in space science, planetary science, astrophysics, or related engineering domains is the primary technical credential.

How JAXA's Interview Format Differs from Other Companies

- **The new graduate process is explicitly standard** — intentionally not special. JAXA's recruitment managers confirm that candidates often ask "are there special examinations?" and "is the interview in English?" — and the answer is no to both. The flow is identical to standard Japanese corporate hiring. Candidates who expect a uniquely demanding or unconventional selection process are frequently surprised by its ordinariness compared to JPL, DLR, or ASI.
- **The aptitude test (/WEB) is a real filter.** The web-based aptitude test covers numerical reasoning, verbal reasoning, and personality assessment — identical in format to most large Japanese employers (SHL, TG-WEB, or equivalent). Performance matters for progression to the interview stage.
- **The field-led career hiring model means the technical interview is conducted by practitioners.** Career hires are evaluated by the actual technical team rather than HR, ensuring specialist expertise is accurately assessed. This means the technical depth probed in a career interview is genuine and domain-specific — an H3 propulsion engineer will be interviewed by JAXA engineers working on LE-9 or SRB-3.
- **Video submission is used as a first selection stage for career hires.** Since 2018, JAXA's career hiring has used video submission as part of the first selection — allowing both candidates and interviewers to engage flexibly before committing to live interview time. The video is not a surprise — candidates prepare it as part of the initial application.
- **The ES (Entry Sheet) is the first written filter for new graduates** — and "why JAXA?" is its most critical question. JAXA's ES asks about research content, motivation for applying, what kind of work you want to do, and your strengths. The motivation section is explicitly where the depth of institutional engagement is assessed.
- **ISAS academic hiring is a wholly separate, university-style process.** Candidates for ISAS positions are selected by the ISAS Advisory Council — not by JAXA HR. The process is closer to a competitive academic job market than to corporate recruitment, with research plan statements, publication lists, and seminar-style presentations.
- **The monozukuri cultural dimension matters less at JAXA than at KHI or MHI.** Unlike JAXA's industrial partners, JAXA itself is a National R&D Agency with a research culture that values scientific output and international recognition alongside engineering manufacturing excellence.

Honest caveat on format variation: JAXA has major sites at Chofu (aeronautics), Tsukuba (satellite engineering and operations), Sagami-hara (ISAS — science), Tanegashima (launch), Uchinoura (small rockets), and Kakuda (propulsion). Each site has a different technical culture and hiring focus. Verify your specific site and division before calibrating your preparation.

PART II: 20 SPECIFIC PREP TASKS

NEW GRADUATE ()

1. Prepare the ES (Entry Sheet) with specific, JAXA-grounded answers — especially the motivation and research sections 4–5 h

Skill: The first written filter and the primary motivation assessment

Why JAXA cares: JAXA's ES is the first selection event and the first assessment of whether a candidate has genuinely engaged with JAXA's mission rather than simply wanting a space-related job. The motivation question specifically probes whether the candidate understands what kind of institution JAXA is and what they would contribute to it. A generic "I love space" answer is insufficient — the expected answer names a specific JAXA programme, connects the candidate's academic background to it, and explains why JAXA's specific role (conducting basic research through development and utilisation) is the right environment for their career.

Done signal: You have written all ES sections, with the motivation answer naming a specific JAXA site and programme area, connecting your academic specialisation to a concrete engineering or research contribution, and containing a sentence that could not be written by a candidate applying to MHI Space or NEC Space. A native Japanese speaker has reviewed all sections for language quality and professional register.

2. Prepare for the WEB aptitude test — numerical, verbal, and personality under timed conditions 4–6 h

Skill: The aptitude test gate that precedes the interview stage

Why JAXA cares: JAXA's aptitude test is standard in format but real as a filter. Candidates who underperform on numerical reasoning or verbal comprehension do not proceed to the interview stage regardless of their research quality. Practising under timed conditions — specifically the format used by the test provider JAXA employs (SHL, TG-WEB, or) — is the highest-leverage early preparation investment.

Done signal: You have completed four full-length practice aptitude tests under timed conditions, scored each, identified your weakest section, and completed targeted practice on that weakness. Your average across three final practice tests is above the 70th percentile.

3. Prepare a research presentation in Japanese — covering your thesis or most significant academic project 3–4 h

Skill: The technical communication element of JAXA's interview rounds

Why JAXA cares: JAXA's technical interviews pay attention to how hard students worked in their academic major and research — the interviewers want to understand the candidate's specific specialty and how deeply they pursued it. The ability to explain your research clearly in Japanese — what problem it addressed, what your specific contribution was, and what it found — is both the technical screen and the language screen simultaneously.

Done signal: You can explain your thesis or most significant project in Japanese in under 5 minutes, covering the engineering or scientific problem, your specific methodological approach, the key result with specific numbers or findings, and one challenge you encountered and resolved. A Japanese engineer has confirmed your technical vocabulary is professional level.

4. Know JAXA's current active missions and programmes — in enough detail to ask an intelligent question about each 5–6 h

Skill: Mission portfolio awareness that enables a genuine conversation with the interview panel

Why JAXA cares: JAXA interviewers distinguish between candidates who want to work in space and those who want to work at JAXA specifically. JAXA's current flagship missions include: H3 rocket (operational from 2024), HTV-X (ISS cargo), Hayabusa2 (ongoing deep space mission), MMX (Phobos sample return, 2026 launch), DESTINY+ (asteroid Phaethon flyby), BepiColombo (Mercury orbit

2026), LUPLEX (lunar polar exploration with ISRO), and Lunar Cruiser (pressurised rover with Toyota for Artemis). Interns who know only Hayabusa2 have not engaged with the breadth of JAXA's current programme.

Done signal: For each of the following — H3 rocket, HTV-X, MMX, DESTINY+, BepiColombo, SLIM legacy/LUPLEX, Lunar Cruiser, and Gateway I-Hab — you can state in Japanese: the mission objective, the current phase, one specific engineering challenge, and one genuine question you would ask the engineering team. You can do this conversationally, not from memorised facts.

5–6 h

5. Understand JAXA's three-founding-organisation structure and identify which site and heritage you are applying to

Skill: Institutional literacy that shapes the specific engineering culture you are entering

Why JAXA cares: JAXA was formed from ISAS (science and planetary exploration), NAL (aeronautics research), and NASDA (satellite development and launch operations). The engineering culture at Sagami-hara/ISAS — where research independence, publication, and scientific curiosity are the primary measures — differs from the culture at Tsukuba — where satellite programme delivery, operational reliability, and ISS mission support are the daily imperatives — which differs from Chofu — where aeronautics research and safety standards govern. A candidate who cannot identify which of these three cultures their target role inhabits is not demonstrating the self-knowledge JAXA's diverse institution requires.

Done signal: You can explain the three founding organisations in Japanese, name the primary site and programme focus of each, state clearly which site you are targeting, explain in one sentence why that specific engineering or research culture aligns with your background, and give one specific example of a JAXA programme at that site that you want to contribute to.

2 h

6. Prepare a specific "JAXA?" answer — that distinguishes JAXA from MHI Space Systems, IHI, NEC, MELCO, and other Japanese space companies

Skill: Company-specific motivation that survives the most common follow-up question

Why JAXA cares: This question is asked in every JAXA interview. The expected answer is not "because JAXA does the most interesting work in Japanese space" — it is a specific explanation of why JAXA's role — conducting integrated operations from basic research through development and utilisation, with an independence that no Japanese aerospace company has — is the right environment for this specific candidate's career aspirations. A candidate who says they want to work at JAXA but whose stated goals would be equally well served at MHI, NEC, or Mitsubishi Electric Space has not answered the question.

Done signal: You have a 90-second answer in Japanese that names a specific JAXA function that is not replicated at industry (basic research, scientific mission design, international programme leadership, or operational independence from commercial constraints), connects your academic background to exercising that function, and explains why the JAXA environment — not an industry partner — enables the contribution you want to make.

2 h

CAREER HIRE (/ 3–10 years)

All new graduate tasks still apply. The research presentation (task 3) must cover professional project work. Mission knowledge (task 4) must extend to engineering depth in the specific programme you would be contributing to.

7. Study the H3 rocket at engineering depth — LE-9 engine, the first-flight failure, and what the successful recovery required

Skill: JAXA's most important current launch vehicle — a programme context every engineering candidate should engage with

Why JAXA cares: The H3 first launch in March 2023 failed when the second-stage engine did not

4 h

ignite. The second launch in February 2024 succeeded, and H3 has subsequently achieved multiple successful operational flights including the HTV-X1 ISS mission in October 2025. Every JAXA engineer involved in this programme — from propulsion to avionics to mission assurance — has lived through both the failure and the recovery. Career hire candidates who can engage with what the second-stage ignition failure investigation required technically, and what validated the corrective action, are demonstrating genuine programme engagement rather than product-brochure knowledge. **Done signal:** You can explain the H3's modular configuration system (first character = number of LE-9 engines, second = number of SRB-3 boosters, S/L = fairing size), what the LE-9 engine uses as its cycle (expander bleed, LOX/LH2), what the hypothesised cause of the first flight second-stage failure was and what the verification approach involved, and what the H3-22S configuration successfully demonstrated in February 2024. You can do this in Japanese. 4 h

8. Prepare a story about working on a spacecraft system — from requirements through test — with a specific anomaly and resolution 2 h

Skill: End-to-end spacecraft development experience for JAXA's engineering track

Why JAXA cares: JAXA's career hiring focus is on immediately deployable specialist expertise — candidates who can contribute to an ongoing programme from early in their tenure. A story about designing, building, and testing a spacecraft subsystem — where an anomaly appeared, the investigation process was rigorous, and the root cause was resolved — demonstrates both the technical competence and the programme culture fit that JAXA needs. Engineers who have only designed in simulation without building and testing hardware are demonstrating a narrower capability.

Done signal: You have a story in Japanese about a spacecraft system or subsystem you contributed to — the design requirement, the test configuration, the anomaly that appeared, the investigation approach, the root cause, and the verified corrective action. The story names the specific spacecraft or programme type, the subsystem, and the failure mode.

9. Understand JAXA's international partnerships — and what they require of JAXA engineers 1–2 h

Skill: International programme interface experience relevant to JAXA's collaborative missions

Why JAXA cares: JAXA is deeply embedded in international programmes — the Gateway I-Hab with ESA, the MMX international collaboration with NASA/CNES/DLR/ESA, LUPEX with ISRO, Lunar Cruiser with Toyota for Artemis, BepiColombo with ESA. JAXA engineers on these programmes must manage technical interfaces in English, navigate multinational programme governance, and represent Japan's technical position credibly in international reviews. JAXA already had significant international teleconference experience before remote work was adopted, making the transition to online international collaboration relatively smooth. Career hire candidates with international programme experience are more immediately deployable in these collaborations.

Done signal: You have a story about working on an international collaborative programme — the partner organisations, the language the interface was conducted in (English for NASA/ESA/CNES/DLR), a specific technical disagreement or ambiguity at the interface, and how it was resolved with formal documentation. The story demonstrates that you can represent your organisation's technical position professionally in English in an international forum.

10. Build technical fluency in your specific spacecraft domain connected to a current JAXA mission 6–8 h

Skill: Domain-specific technical depth that demonstrates immediate deployability

Why JAXA cares: Career hires at JAXA are evaluated by the technical team that will work with them — the evaluation is specifically about whether the candidate's expertise fills a gap the team has right now. The most deployable career hires are those whose specific technical background maps to a current programme need: a thermal engineer for MMX's Phobos thermal environment, a GNC engineer

for DESTINY+’s ion propulsion navigation, an RF engineer for deep space communication on H3 or HTV-X. Generic spacecraft engineering capability is less compelling than focused domain depth.

Done signal: For the specific JAXA mission or programme most relevant to your technical background, you can explain: the mission’s current phase, the specific technical challenge in your domain that the mission faces, the approach JAXA has taken (or is considering) to address it, and how your specific professional experience contributes to or improves on that approach. You can answer three follow-up questions on each without going vague.

6–8 h

11. Study the MMX mission at engineering depth — sample return from Phobos and the international collaboration structure

Skill: JAXA’s most ambitious current planetary exploration programme

Why JAXA cares: MMX is planned to launch on H3 from Tanegashima in FY2026, orbit Phobos for three years, collect samples with the IDEFIX rover deployed on the surface, and return samples to Earth around 2031. It is an international project led by JAXA with NASA, CNES, DLR, and ESA providing instruments and ground support — the largest international collaborative mission in lunar and planetary exploration ever led by Japan. Career engineers interviewing for roles on MMX or related planetary programmes who cannot describe the scientific objectives, the sample return architecture, the international workshare, or the primary engineering challenge are demonstrating they have not engaged with the programme they are applying to support.

Done signal: You can explain MMX’s scientific objectives (origin of Martian moons — capture vs. giant impact hypothesis), the mission architecture (IDEFIX rover deployment, surface sampling, three-year Phobos orbit, Earth return via Australian recovery), the technology heritage from Hayabusa2, the four international partner contributions (NASA’s MEGANE spectrometer, CNES/MIRS spectrometer, DLR’s DESTINY Dust Analyzer, ESA’s deep space communication equipment), and one specific engineering challenge in sampling a low-gravity, unknown-composition surface.

4 h

12. Prepare a story about a programme failure or anomaly you investigated — specifically in a space or aerospace context

Skill: Anomaly investigation culture specific to JAXA’s mission-critical heritage

Why JAXA cares: JAXA’s institutional memory includes several significant anomalies — the Hayabusa1 multiple system failures that nearly prevented the world’s first asteroid sample return, the H-IIA F6 loss of ALOS-3, and SLIM’s tilted landing that surprisingly enabled solar panel power. The culture built on these experiences is one of rigorous anomaly investigation, transparent reporting, and systematic corrective action. Career engineers who have conducted formal anomaly investigations — root cause analysis, failure mode documentation, corrective action verification — are embodying the engineering culture JAXA values.

Done signal: You have a story about a spacecraft or launch system anomaly you investigated — the anomaly symptom, the investigation methodology, the false hypotheses that were eliminated, the confirmed root cause, the corrective action, and the verification that closed the investigation. The story demonstrates systematic thinking rather than intuitive guessing.

1–2 h

13. Understand JAXA’s role in the Artemis programme and Lunar Gateway — Japan’s specific commitments

Skill: Lunar exploration programme context for JAXA’s most strategically significant current international commitment

Why JAXA cares: Japan’s participation in the Artemis programme is extensive: HTV-X will deliver cargo to Gateway, JAXA is contributing to the Gateway I-Hab module with ESA, the Lunar Cruiser pressurised rover is being developed with Toyota, and Japan signed the Artemis Accords and a bilateral lunar surface exploration agreement with NASA in 2024. This commitment is JAXA’s most

3 h

3 h

significant international alignment decision since joining the ISS programme — and engineers on Gateway, HTV-X, or lunar exploration programmes must understand the strategic context, the bilateral agreement structure, and what Japan's specific workshare involves.

Done signal: You can explain Japan's Artemis programme contributions (HTV-X to Gateway, I-Hab with ESA, Lunar Cruiser with Toyota, Japanese astronaut lunar landing agreement with NASA), what the bilateral NASA-Japan lunar surface exploration agreement covers, what Gateway's orbit (NRHO — near-rectilinear halo orbit) implies for HTV-X docking compared to the ISS, and one specific engineering challenge in operating a pressurised rover in permanently shadowed polar regions.

ISAS ACADEMIC STAFF / SENIOR RESEARCHER (/)

All career hire tasks still apply. The publication record must be substantive and show a coherent research programme. The international partnership story (task 9) must include experience representing Japan's scientific position in a multinational mission review. The anomaly investigation story (task 12) must be from a mission or campaign you personally led the investigation for.

6–8 h

14. Prepare to present your research programme — the central scientific question, your contributions, and your next five years at JAXA

Skill: The academic seminar-style presentation that is ISAS hiring's primary evaluation event

Why JAXA cares: ISAS positions are selected through an academic-style process overseen by the ISAS Advisory Council — a research plan statement, publication list, and interview. The hiring decision is a bet on a research direction — the institute is investing in someone who will advance a specific frontier of space science or planetary exploration for the duration of their tenure. Candidates who can articulate a coherent scientific programme — the question that motivates it, the methods that advance it, the JAXA missions it would enable or exploit — are making the strongest possible case.

Done signal: You can give a 15-minute research programme presentation (in Japanese or English depending on the position) covering: the central scientific question, the current state of knowledge, your specific contributions to date and their significance, the next 3–5 investigations you would pursue at JAXA, the specific JAXA missions or infrastructure you would use (ISAS facilities, space mission data, or future mission proposals), and one unsolved problem that drives your continued work.

2 h

15. Prepare specific examples of competing for and winning research funding — connected to JAXA's Drittmittel-equivalent environment

Skill: Competitive grant acquisition as a senior research function at a National R&D Agency

Why JAXA cares: JAXA operates in an environment where a significant fraction of research is funded through competitive grants — JSPS (Japan Society for the Promotion of Science), MEXT (Ministry of Education, Culture, Sports, Science and Technology), JST (Japan Science and Technology Agency), and international sources (ESA, NASA bilateral grants). Senior researchers who can attract competitive funding are enabling their own research programmes and demonstrating that the scientific community validates their research agenda. The ability to articulate a successful funding narrative — what problem the proposal addressed, how reviewers were convinced — is a senior research capability.

Done signal: You have a story about a competitive research grant you led or co-led — the funding source, the scientific question, the review process, the feedback received, and whether it was funded. You can also articulate one specific competitive funding opportunity you would pursue at JAXA and why your scientific background positions you to win it.

16. Prepare a technically grounded position on the scientific questions MMX will answer — and what the samples from Phobos will uniquely enable 4 h

Skill: Planetary science depth at the frontier of JAXA's most ambitious current mission

Why JAXA cares: MMX's primary scientific goal is to determine whether Phobos is a captured asteroid or a fragment from a giant impact on early Mars — answering a fundamental question about the formation of the Martian system and the supply of water and organic material in the inner solar system. ISAS senior scientists are expected to have a technically grounded view on what the Phobos samples will contain, what analysis techniques will discriminate between the formation hypotheses, and what JAXA's sample curation and distribution plans should ensure. Scientists who can engage with this at a peer level — rather than describing the mission's objectives from a press release — are having the right conversation.

Done signal: You have a prepared 3-minute technical discussion in Japanese (or English for international positions) of one specific scientific question MMX addresses — what measurement distinguishes the capture hypothesis from the impact hypothesis, what isotopic ratio or mineral assemblage would be diagnostic, what the Ryugu sample analysis taught us about Hayabusa2's sample handling that informs MMX's approach, or what the IDEFIX rover surface traverse needs to accomplish to ensure sample site representativeness.

17. Prepare specific examples of supervising graduate students or junior researchers — with outcomes that demonstrate independence 1–2 h

Skill: Academic mentorship as a JAXA/ISAS senior research function

Why JAXA cares: ISAS is embedded in Japan's academic ecosystem — it supervises doctoral students in partnership with universities, hosts postdoctoral researchers, and is expected to contribute to Japan's scientific workforce development. Senior ISAS staff are expected to have supervised researchers who now independently contribute to the field. The quality of this supervision — whether you developed research independence or maintained dependency — is directly relevant to your institutional value.

Done signal: You have a story about supervising a graduate student or postdoctoral researcher — their research question, your approach to developing their independent research capability, a specific moment when you shifted from directing to advising, the research output that demonstrated their independence, and their subsequent career position. The story demonstrates that your supervision produced an independent scientist.

18. Build quantitative estimation fluency for spacecraft mission parameters — specific to your scientific domain 4 h

Skill: Mission design first-principles reasoning at senior scientific speed

Why JAXA cares: Senior ISAS scientists are expected to evaluate new mission concepts, assess feasibility against cost and mass constraints, and respond credibly to "does this number make sense?" in programme reviews. For a planetary scientist: estimating orbital delta-v requirements from a pork-chop plot, estimating instrument SNR from aperture and integration time, estimating science data return from link budget assumptions. None of these requires full simulation to estimate at order-of-magnitude if the fundamentals are solid.

Done signal: You have worked through five mission analysis or instrument design estimation problems relevant to your specific scientific domain — stated out loud in Japanese, assumptions named before numbers — arriving at order-of-magnitude answers with stated uncertainty. For each, you have identified the most load-bearing assumption and how you would verify it.

ALL LEVELS

19. Prepare a specific answer to "/" (What kind of engineer/researcher do you want to become?) — grounded in JAXA's specific mission context

1-2 h

Skill: Career aspiration articulation that connects to JAXA's dual identity as engineering institution and science institute

Why JAXA cares: This question appears in JAXA's ES and in every interview round. The expected answer at JAXA — unlike at MHI or NEC Space — must engage with the specific nature of working at a national R&D agency where engineering serves science and science motivates engineering. The engineer who wants to build satellites that address unsolved planetary science questions, or the scientist who wants to design missions that only JAXA's integration of research and development enables, is describing something specifically available at JAXA and not at Japan's industrial partners. Generic aspirations about "advancing space technology" are not differentiated.

Done signal: You have a 90-second answer in Japanese that names a specific type of technical mastery (H3 propulsion system reliability, deep space navigation for small body missions, planetary surface sampling robotics, or equivalent), connects it to a specific JAXA mission where that mastery is needed, and frames the aspiration in terms of what Japan specifically needs to contribute to international space science — not just personal career advancement. The answer could not be given unchanged to MHI or NEC.

20. Verify Japanese language proficiency and site location flexibility — and understand what both mean for your specific target role

1-2 h

Skill: Eligibility and practical self-knowledge for a national organisation with multiple distinct sites

Why JAXA cares: JAXA's technical staff are based across multiple sites — Tsukuba (satellite engineering), Sagami-hara/ISAS (science), Chofu (aeronautics), Tanegashima (launch operations), Uchinoura (small rockets), and Kakuda (propulsion). Unlike private sector employers where Tokyo proximity is often assumed, JAXA's roles require genuine location flexibility — accepting assignment to Tanegashima or Uchinoura is a real career commitment. Additionally, the working language at all domestic sites is Japanese, and interview questions about relocation willingness and Japanese proficiency are standard. Candidates who have not honestly assessed either factor will be surprised after joining.

Done signal: You know your Japanese proficiency level (JLPT grade or equivalent honest self-assessment), have reviewed the specific site location requirements for your target role, and can answer the recruiter's questions about language and relocation willingness clearly and without hesitation. If any site assignment would be genuinely difficult for personal reasons, you have identified which roles remain accessible and calibrated your application accordingly.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	ES (Entry Sheet) — JAXA-specific motivation and research sections	All	4–5 h
2	WEB aptitude test — timed practice, all sub-types	All	4–6 h
3	Research presentation in Japanese — 5 min, technical depth	All	3–4 h
4	Active mission portfolio — H3, HTV-X, MMX, DESTINY+, Lunar Cruiser	All	5–6 h
5	Three-founding-organisation structure — identify your target culture	All	2 h
6	"JAXA?" — distinct from MHI/NEC/MELCO, programme-specific	All	2 h
7	H3 rocket engineering depth — LE-9, first-flight failure and recovery	Mid+	4 h
8	Spacecraft system development story — anomaly and resolution	Mid+	2 h
9	International programme interface story — in English	Mid+	1–2 h
10	Domain technical depth — connected to specific current JAXA mission	Mid+	6–8 h
11	MMX — sample return architecture, international structure, science	Mid+	4 h
12	Anomaly investigation story — systematic, space/aerospace context	Mid+	1–2 h
13	Artemis programme — Japan's Gateway, HTV-X, Lunar Cruiser role	Mid+	3 h
14	Research programme presentation — 15 min, ISAS seminar format	Senior/ISAS	6–8 h
15	Competitive funding acquisition — grant story and next proposal	Senior/ISAS	2 h
16	MMX scientific questions — Phobos formation hypothesis discrimination	Senior/ISAS	4 h
17	Graduate student/postdoc supervision — independence outcome	Senior/ISAS	1–2 h
18	Mission analysis quantitative estimation — five problems in Japanese	Senior/ISAS	4 h
19	"/?" — JAXA-specific aspiration	All	1–2 h
20	Japanese language level and site relocation flexibility verification	All	1–2 h

THE STRUCTURAL REALITY

The structural reality that makes JAXA unlike every other organisation in this series: JAXA is the only organisation in this guide that has achieved the world's first asteroid sample return — twice — and is now planning the first sample return from a moon of another planet, while simultaneously operating Japan's independent access to space through a newly operational launch vehicle, contributing a human habitat module to a Moon-orbiting space station, and developing a pressurised rover for the lunar surface with an automobile manufacturer.

This breadth of mission — from basic planetary science through launch vehicle engineering through human spaceflight — at a single institution with roughly 3,000 employees, is exceptional anywhere in the world. It is available because JAXA's founding merger preserved three distinct scientific and engineering communities within one institution, and because Japan's space policy has consistently chosen to fund a genuinely integrated national space capability rather than splitting the basic research, development, and operations functions across separate organisations.

The engineers and scientists who succeed at JAXA are those who find meaning specifically in this integrated identity — who want to work where the scientific question motivates the engineering mission and the engineering capability enables the scientific discovery — and who understand that this integration, conducted at Japanese engineering standards with Japanese methodological rigour, is the institutional character that has put Japan's name on the first asteroid samples, the first precision lunar landing, and the first Martian moon mission.

BREAK INTO AEROSPACE